Data management plan (DMP)

Frost Day Prediction in Vienna

FDPV

Version	Effective date	Description of document/changes
1.0	30/05/2026	First version of the DMP – created for the start of the project

Level of distribution		

FWF Data Management Plan (DMP)

I General Information	
I.1 Administrative information	**Internal Project ID:** DaST-2026 **DMP version:** 1.0, 30.05.2026 **Contributors:** Vivek Sharma, ORCID: 0009-0006-4879-6388 Sophie Konecny, ORCID: 0009-0006-5745-5729 Anxhela Sulmina, ORCID: 0009-0008-9371-0488 Nayma Alam, ORCID: 0009-0003-4731-9553
I.2 Data management responsibilities and resources	**Person responsible for data management and DMP:** Anxhela Sulmina **Co-ordination of data management responsibilities across partners:** Sophie Konecny, Vivek Sharma, Nayma Alam **Resources costed in for RDM:** There are no costs dedicated to data management and ensuring that data will be FAIR.
II Data Characteristics	
II.1 Data description and collection or re-use of existing data	**Produced datasets:**

dataset ID	title	type	format	estimated volume	contains sensitive data	description
P1	Frost Day Prediction Random Forest Model	Software applications	joblib	3.6 MB	no	A Random Forest Classifier trained on daily meteorological measurements from Vienna Hohe Warte station (2000-2023) to predict frost days (minimum temperature below 0°C). The model is serialised in joblib format using scikit-learn. It includes

						hyperparameters tuned via 5-fold cross-validation and achieves accuracy of 97.1% and ROC-AUC of 0.995 on the held-out test set. Deposited in the TU Wien Research Data Repository with a persistent DOI.
P2	Frost Day Prediction Evaluation Results	Structured text	CSV, PNG	< 10 MB	no	Evaluation artefacts produced by the frost day prediction experiment including: confusion matrix (PNG), ROC-AUC curve (PNG), feature importance chart (PNG), per-feature histograms (PNG), correlation heatmap (PNG), and evaluation metrics (CSV) containing accuracy, precision, recall, F1-score and ROC-AUC on the held-out test set. All figures are saved in PNG format. Deposited in the TU Wien Research Data Repository as a separate deposit from the model, with a persistent DOI.

Reused datasets:

dataset ID	title	source	rights (e.g. license)	contains sensitive data	description
R1	Messstationen Tagesdaten v2 (Daily Station Measurements)	https://dataset.api.hub.geosphere.at/v1/station/historical/klima-v1-1d	CC0	no	…

Methods and software used for data generation and reuse

The input dataset is reused from GeoSphere Austria Data Hub (Messstationen Tagesdaten v2), accessed via the public REST API (https://dataset.api.hub.geosphere.at/v1/station/historical/klima-v1-1d). The data is downloaded in CSV format and loaded into DBRepo using the DBRepo Python REST API client. The experiment is implemented as a Python 3.11 pipeline following the CRISP-DM methodology. The following software and

	libraries are used: scikit-learn for Random Forest classification and model evaluation, pandas and NumPy for data preparation and feature engineering, Matplotlib and Seaborn for visualisation, and joblib for model serialisation. The trained model, evaluation figures and metrics are stored as output artefacts and deposited in the TU Wien Research Data Repository with persistent identifiers.
III Documentation and Data Quality	
III.1 Metadata and documentation	**Data organisation, metadata, and documentation:** The input data is stored in a relational database in DBRepo following a Third Normal Form (3NF) schema consisting of two tables: station (station metadata) and daily_observation (daily meteorological measurements). The schema includes primary and foreign key constraints to ensure data integrity. All code, notebooks, and configuration files are version-controlled in a public GitHub repository using Git. A consistent file-naming convention is applied covering input datasets, output files, scripts and configuration files, documented in the README. GitHub releases are used to create versioned snapshots of the experiment at key milestones. The repository is connected to Zenodo for automatic DOI minting upon each release, ensuring persistent and citable versions of the software. Output artefacts including the trained model, evaluation figures and metrics are deposited in the TU Wien Research Data Repository as separate versioned deposits with full metadata. To make the project easier to understand and reuse, we document the data, software, and machine learning outputs using several established metadata standards. Meteorological variables are linked to CF Standard Names, while observation and station information is described using SOSA/SSN concepts. Units of measurement are documented using the SI Digital Framework ontology and stored in DBRepo metadata. The project is also described using RO-Crate, CodeMeta, FAIR4ML, and a Model Card. Together, these standards document the software environment, dependencies, model characteristics, evaluation results, and relationships between project artefacts. Dataset descriptions are additionally provided using a Croissant JSON-LD record. All deposits in the TU Wien Research Data Repository include metadata such as title, description, keywords, creators, licences, and persistent identifiers. A CITATION.cff file is included in the repository to support proper citation of the project outputs. Additionally, we will provide common metadata such as title, description, or keywords when publishing data in open access repositories. In such a case, we will follow the default template provided by the repository, such as Data Cite Metadata or Dublin Core. As far as possible, we will use controlled vocabularies for our data to allow inter-disciplinary interoperability and machine-actionability. The experiment follows the CRISP-DM methodology and is fully documented to ensure reproducibility and reuse. All data processing, model training and evaluation steps are implemented in Jupyter notebooks stored in the src/ folder of the GitHub repository. The notebooks include inline comments explaining each step and decision. A step-by-step reproduction guide is provided in the README.md covering requirements, installation, and execution instructions. The relational database schema is documented with an Entity-Relationship diagram and SQL CREATE statements stored in the repository. All DBRepo tables include descriptive column metadata, semantic concept mappings and unit of measurement mappings via ontology URIs. The trained model is accompanied by a Model Card (docs/model-card.md) documenting training data, evaluation metrics, intended use, limitations and ethical considerations. FAIR4ML metadata records all hyperparameters, evaluation metrics and algorithm details. A standards overlap analysis is included in the final report comparing RO-Crate, CodeMeta, FAIR4ML, Croissant and Model Card. Fixed random seeds and stratified splits ensure that the experiment produces identical results when re-executed. The API-based data loading reimplementation (T2.6) is verified to produce results identical to the original local-file version.

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<td>III.2 Data quality control</td>
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Data quality control:
The input data is provided by GeoSphere Austria, the national meteorological service, and is considered a reliable source of weather observations. During data preparation, missing values are identified and handled appropriately, invalid or incomplete records are removed, and data types are verified before loading the data into DBRepo.
The database schema includes primary and foreign key constraints to maintain consistency between tables. For the machine learning workflow, a fixed random seed and a stratified train/validation/test split are used to ensure reproducible results. Model performance is evaluated using accuracy, precision, recall, F1-score, and ROC-AUC. We also verified that the API-based implementation produces the same results as the original local-file version.
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<td colspan="2">IV Data Storage, Sharing, and Long-Term Preservation</td>
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<td>IV.1 Data storage and backup during the research process</td>
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Storage and backup facilities:
For the duration of the project, storage and backup of data will be ensured by Anxhela Sulmina (acting as the person responsible for data management and DMP) in cooperation with the system operator. The data will not be stored on the servers of TU Wien.
Data security and protection of sensitive data:
We pay strict attention to compliance with the relevant institutional and national data protection policies. At this stage, it is not foreseen to process any sensitive data in the project. If this changes, advice will be sought from the data protection specialist at TU Wien, and the DMP will be updated.
Access to data during research:
<table>
<tr><th>dataset ID</th><th>selected project members</th><th>all other project members</th><th>the public</th></tr>
<tr><td>P1</td><td>writing</td><td>reading only</td><td>reading only</td></tr>
<tr><td>P2</td><td>writing</td><td>reading only</td><td>reading only</td></tr>
<tr><td>R1</td><td>writing</td><td>reading only</td><td>reading only</td></tr>
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<td>IV.2 Data sharing and long-term preservation</td>
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Data publication and access conditions:
As far as possible, obtained datasets will be published in repositories. Details on access conditions, reuse licenses, reasons for restrictions, etc. are collected in the table below.
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dataset ID	access conditions	estimated publication date	location for publication (repository)	PID	license
P1	Open	2026-05-30	TU Wien Research Data	https://doi.org/10.70124/th8xh-52s95	CC-BY-4.0
P2	Open	2026-05-30	TU Wien Research Data	https://doi.org/10.70124/8t7w0-5pt49	CC-BY-4.0

Repository description:

TU Wien Research Data is an institutional repository of TU Wien to enable storing, sharing and publishing of digital objects, in particular research data. It facilitates the funders' requirements for open access to research data and the FAIR principles by making research output findable, accessible, interoperable, and reusable. A DOI is assigned to each dataset published in TU Wien Research Data. This service is developed by the TU Wien Center for Research Data Management and hosted by TU.it. https://researchdata.tuwien.ac.at/

Methods or software needed to access and use data: To access the input data, users require a standard HTTP client or web browser to query the GeoSphere Austria Data Hub API or the DBRepo REST API. No authentication is required for either. To reuse the experiment code, users need Python 3.11 or higher with the following libraries: scikit-learn, pandas, NumPy, Matplotlib, Seaborn, joblib, and the dbrepo Python client. All dependencies are documented in the repository with version pins in requirements.txt. The trained model is serialised in joblib format, which is compatible with any scikit-learn installation. Step-by-step reproduction instructions are provided in the README.md.

Long-term preservation and deletion of data:

dataset ID	location for long-term storage	minimum retention period (≥ 10 years)	foreseeable research uses and/or users
P1	TU Wien Research Data	10 years	The primary target audience includes meteorologists, climate scientists, and data scientists interested in weather prediction and frost day analysis for the Vienna region. The dataset and experiment may also be of interest to urban planners, agricultural researchers, and public health professionals who need to anticipate frost events. Additionally, educators and students in data science and machine learning courses may find the experiment useful as a reproducible example of a FAIR-compliant open-science workflow. The CC0 licence on the input data and MIT licence on the code place no restrictions on reuse.
P2	TU Wien Research Data	10 years	

V Legal and Ethical Aspects	
V.1 Legal aspects	**Personal data:** At this stage, it is not foreseen to process any personal data in the project. If this changes, advice will be sought from the data protection specialist at TU Wien, and the DMP will be updated. **Intellectual property rights and rights of use:** The following individual(s) hold rights and control access to the project data: All datasets and research outputs are publicly accessible under open licences. No access restrictions are applied. The GitHub repository is owned by the group account datastewgroup4. All four group members (Role A, B, C, D) have collaborator access with write permissions to the repository. The DBRepo database (ID: a16a980a-3489-492b-adcf-74c70a07248b) is owned by the group account datastewgroup4. All group members have write access to the database. The TU Wien Research Data Repository deposits are created by the group members with all four listed as creators. The deposits are publicly accessible upon publication. The Zenodo deposit is connected to the GitHub repository via the GitHub-Zenodo integration and is publicly accessible. After the project concludes, the TU Wien Research Data Repository and Zenodo will serve as the long-term archives, maintained by their respective institutions.
V.2 Ethical aspects	**Ethical issues:** No particular ethical issue is foreseen with the data to be used or produced by the project. This section will be updated if issues arise.